

BINOMIAL

THEROM ...

SUB-TOPICS

- * Factorial notation.
- * Binomial theorem
- * General term for
 - $\Rightarrow (a+b)^n$
 - $\Rightarrow (a-b)^n$
- * Greatest binomial coefficient
- * Numerically greatest term. (JEE Adv Pqs)
- * Concept of Middle term.
When n is even, odd
- * No. of rational term.
- * Term independent on x .
- * n^{th} term from Beginning and end.
- * Sum, Product of Binomial coeff & their Properties
- * Method of finding Sum of coeff:

$\rightarrow$ Sigma notation.

$\rightarrow$ Differentiation

$\rightarrow$ Integration.

Max Time we get

$$r!(r+1) = (r+1)!$$

• Multiple & Divide By $\frac{n+1}{n+1}$

$$\frac{(n+1)!}{(n+1)!(r+1)!}$$

$$\frac{(n+1)!}{(n+1)!(r+1)!} \rightarrow {}^{n+1}C_{r+1}$$

idea is
Adding & sub.

- * Middle term.
- * No. of terms, General Exp. ...
- * Double summation.
- * Remainder theorem.
- * last n digits figure out
- * Binomial theorem for Any index.

Tricks Key Points :

Formulas :

$$\begin{aligned} * \sum_{r=1}^n r {}^n C_r &= n \times 2^{n-1} \\ * \sum_{r=1}^n r^2 {}^n C_r &= n(n+1) \cdot 2^{n-2} \\ * \sum_{r=0}^n \frac{{}^n C_r}{r+1} &= \frac{2^{n+1}-1}{n+1} \\ * \sum_{r=0}^{n-k} {}^n C_r \times {}^n C_{r+k} &= {}^{2n} C_{n-k} \\ * \sum_{r=1}^n {}^n C_r &= 2^n - 1 \end{aligned}$$

Binomial Theorem:

$$(a+b)^n = {}^n C_0 a^n + {}^n C_1 a^{n-1} b + {}^n C_2 a^{n-2} b^2 + \dots + {}^n C_n b^n$$

$$(a-b)^n = {}^n C_0 a^n - {}^n C_1 a^{n-1} b + \dots$$

$$T_{r+1} = {}^n C_r a^{n-r} b^r$$

$$T_{r+1} = (-1)^r {}^n C_r a^{n-r} b^r$$

$$\Rightarrow {}^n C_r = \frac{n!}{r!(n-r)!}$$

$$r \leq \frac{(n+1)|b|}{|a|+|b|}$$

$$r \geq \frac{n|b|-|a|}{|a|+|b|}$$

For n is distinct

Results:

$$i. {}^n C_0 = \frac{n!}{n!} = 1$$

$$ii. {}^n C_n = 1$$

$$iii. {}^n C_r = {}^n C_{n-r}$$

$$iv. {}^n C_r = {}^n C_s \quad |r=s| \text{ or } |n=r+s|$$

$$v. {}^n C_r + {}^n C_{r-1} = {}^{n+1} C_r$$

$$vi. \frac{{}^n C_r}{{}^n C_{r-1}} = \frac{n-r+1}{r}$$

$$vii. {}^n C_r = \frac{n}{r} \times (n-1) C_{r-1} = \frac{n(n-1)}{r(r-1)} \times {}^{n-2} C_{r-2}$$

- K^{th} term from end is equal to $(n-K+1)^{\text{th}}$ term from the beginning

⇒ n^{th} term from Beginning of $(a+b)^n$ is equal to n^{th} term from Ending ... $(b+a)^n$

⇒ If n^{th} term given $\boxed{r=n-1}$

⇒ Condition:

$$6({}^nC_6) = nC_{n-6}$$

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Properties of Binomial:

i. ${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n = 2^n$

ii. ${}^nC_0 - {}^nC_1 + {}^nC_2 - {}^nC_3 + \dots = 0$

iii. ${}^nC_0 + {}^nC_2 + {}^nC_4 + \dots = 2^{n-1}$

iv. ${}^nC_1 + {}^nC_3 + {}^nC_5 + \dots = 2^{n-1}$

Product:

$${}^nC_0 {}^nC_r + {}^nC_1 {}^nC_{r+1} + {}^nC_2 {}^nC_{r+2} + \dots + {}^nC_{n-r} {}^nC_n = \frac{(2n)!}{(n+r)!(n-r)!}$$

Double Summation:

i. $\sum_{0 \leq i \leq j \leq n} {}^nC_i {}^nC_j = 2^n {}^nC_n$

ii. $\sum_{0 \leq i} \sum_{j \leq n} {}^nC_i {}^nC_j = 2^{2n} = 4^n$

iii. $\sum_{0 \leq i < j \leq n} {}^nC_i {}^nC_j = \frac{4^n - 2^n {}^nC_n}{2}$

iv. $\sum_{j=0}^n \sum_{i=0}^j {}^nC_i {}^nC_j = 2^{2n}$

Approximations:

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{1 \cdot 2} x^2 + \frac{n(n-1)(n-2)}{1 \cdot 2 \cdot 3} x^3 + \dots$$

Formulas

$$(1+x)^n (1+x)^n = (nC_0 + nC_1 x + nC_2 x^2 + \dots) (nC_0 + nC_1 x + nC_2 x^2 + \dots)$$

$$(1+x)^{2n} \Rightarrow 2nC_{n-1} = (nC_0 nC_1 + nC_1 nC_2 + \dots + \dots)$$

$$1. nC_0 nC_r + nC_1 nC_{r+1} + \dots + nC_{n-r} nC_n = 2nC_{n-r}$$

$$2. nC_0^2 + nC_1^2 + nC_2^2 + nC_3^2 + \dots + nC_n^2 = 2nC_n$$

$$3. nC_0 nC_1 + nC_1 nC_2 + nC_2 nC_3 + \dots + nC_{n-1} nC_n = 2nC_{n-1}$$

$$4. nC_0 mC_r + nC_1 mC_{r+1} + nC_2 mC_{r+2} + \dots = m+nC_{m-r}^{m-r}$$

$$5. nC_0 mC_r + nC_1 mC_{r-1} + nC_2 mC_{r-2} + \dots + nC_r mC_0 = m+nC_r$$

Note:

$$nC_0^2 - nC_1^2 + nC_2^2 - \dots + (-1)^n nC_n^2 = \begin{cases} 0 & \text{if } n \text{ is odd} \\ (-1)^{\frac{n}{2}} nC_{\frac{n}{2}} & \text{if } n \text{ is even} \end{cases}$$

- Sum of binomial coeff
- Sum of irrational rational
- NO. of irrational rational terms.
- Middle terms
- coeff of Particular term
- Greatest Term
- Constant term

Remainder theorem :

☐ a^b is Divisible by c then K is the remainder $0 \leq K < c$, $K \in \mathbb{W}$.

Try to express a as $(\text{Multiple of } c) + P$ $P < c$

and use binomial theorem to reduce it !

1Q. find remainder when 2^{32} is Divided by

i. 5.

ii. 11.

iii. 10.

$$\begin{aligned} \text{Sol: } 2^{32} &= (2^4)^8 = (16)^8 \\ &= (15+1)^8 \\ &= (15)^8 + 8C_1 15^7 + \dots + 8C_7 15 + 8C_8 \\ &= 15N + 1 \\ &= 15P + 1 \end{aligned}$$

Remainder

is (1).

$$\begin{aligned} \text{ii } 11 : \frac{2^{32}}{11} &= (2^2)(2^{30}) \\ &= 4(2^6)^5 \\ &= 4(32)^5 \\ &= 4(33-1)^5 \\ &= 4[6C_0(33)^5 - 6C_1(33)^4 + \dots + 6C_5(1)^5] \\ &= 4(11K) + 4(1) \\ &= 4. \end{aligned}$$

2Q. The remainder when 5^{99} is divided by 13 —

$$\begin{aligned} 5 \cdot (5)^{98} &= 5(26-1)^{49} = 5[(49C_0(26)^{49} - 49C_1(26)^{48} + \dots + 49C_{48}(1)^{49} \\ &\quad - 49C_{49}(1)^{49}] \end{aligned}$$

$$= \frac{(2 - \frac{49}{6}x)(1 - \frac{1}{18}x)}{3}$$

$$= \frac{(2 - \frac{1}{9}x - \frac{49}{6}x + \frac{49}{6 \times 18}x)}{3}$$

* *

* Multinomial Theorem:-

$$(x_1 + x_2 + x_3 + \dots + x_r)^n = \frac{n!}{a_1! a_2! \dots a_r!} x_1^{a_1} x_2^{a_2} \dots x_r^{a_r}$$

$$a_1 + a_2 + \dots + a_r = n, \quad 0 \leq a_1, a_2, \dots, a_r \leq n$$

ex: find coefficient of $a^2 b^3 c^4$ in $(a+b+c)^9$.

$$\Rightarrow \frac{9!}{2!3!4!} (a)(b)(c)$$

$$\Rightarrow \frac{9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 3 \times 2 \times 1 \times 4 \times 3 \times 2 \times 1}$$

$$\Rightarrow 45 \times 7 \times 4$$

$$\Rightarrow 45 \times 28$$

$$\Rightarrow 1260$$

$$\begin{array}{r} 4 \\ 45 \\ \hline 28 \\ 360 \\ 90+ \\ \hline 1260 \end{array}$$

ex: find coeff of $a^3 b^4 c^5$ in $(2a+3b-4c)^{12}$

$$\Rightarrow (2a+3b-4c)^{12}$$

$$\Rightarrow \frac{12!}{(2a)^3 (3b)^4 (-4c)^5} (2a)^3 (3b)^4 (-4c)^5$$

$$\Rightarrow \frac{12 \times 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{3!4!5!} 8a \cdot 81b^4 \cdot (-4)^5 c^5$$

$$\Rightarrow \frac{12!}{3!4!5!} \cdot 8 \cdot 81 \cdot (-4)^5 a^3 b^4 c^5$$

$$\left| \frac{4x}{5} \right| < 1$$

$$\frac{4}{5}|x| < 1 \quad \therefore |x| < \frac{5}{4}$$

$$-\frac{x}{4} < x < \frac{5}{4}$$

• Only in this interval eq (1) exp is possible.

if $|t| < \frac{1}{2}$ and 4th term in exp of $(1-2t)^{-3/4}$ is $\frac{7!t^3}{K}$ $K=?$

Ans: $\Rightarrow (1-2t)^{-3/4}$

$$\Rightarrow 1 + \frac{(-3/4)(-2t)}{1!} + \frac{(-3/4)(-3/4-1)(-2t)^2}{2!} + \frac{(-3/4)(-3/4-1)(-3/4-2)(-2t)^3}{3!} + \dots$$

$$+ \frac{3 \times 7!}{128 \times 16} t^3$$

$$\Rightarrow \boxed{K=16}$$

Special Case:

When n is negative integer.

let $n = -m$ where $m \in \mathbb{N}$

$$(1+x)^{-m} = 1 - mC_1 x + \frac{m(m+1)}{2!} x^2 - \frac{m(m+1)(m+2)}{3!} x^3 + \dots$$

$$(1+x)^{-m} = 1 - \frac{m}{1!} x + \frac{(m+1)(m+2)}{2!} x^2 - \dots$$

PUT $n = -m$

in General term.

$$\Rightarrow (1+x)^{-m} = 1 - mC_1 x + \frac{(m+1)(m+2)}{2!} x^2 - \frac{(m+2)(m+3)}{3!} x^3 + \dots$$

General term $T_{r+1} = (-1)^r \frac{(m+r-1)!}{r!} x^r$

$$\Rightarrow (1-x)^{-m} = 1 + mC_1 x + \frac{m(m+1)}{2!} x^2 + \frac{m(m+1)(m+2)}{3!} x^3 + \dots$$

General term $T_{r+1} = \frac{(m+r-1)!}{r!} x^r$

Ex: $(1-x)^{-1} = 1 + x + x^2 + x^3 + \dots$

Find coeff of t^4 in $\left(\frac{1-t^6}{1-t} \right)^3$ is 15

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$$(1-t^6)^3 (1-t)^{-3} \Rightarrow (3C_0 - 3C_1 t^6 + 3C_2 t^{12} - t^{18}) (1-t)^{-3}$$

$$\Rightarrow (3C_0 - 3C_1 t^6 + 3C_2 t^{12} - t^{18}) (1 + 3C_1 t + 6C_2 t^2 + 10C_3 t^3 + \dots)$$

Binomial Theorem for Any index :

Binomial Theorem for $n \in \mathbb{Q}^+$.

* * IF $n \in \mathbb{Q}^+$ and $|x| < 1$ then

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

$$(1+x)^{\frac{p}{q}} = 1 + \frac{p}{q}x + \frac{\frac{p}{q}(\frac{p}{q}-1)}{2!}x^2 + \dots$$

$$(1+x)^{-n} = 1 - nx + \frac{n(n+1)}{2!}x^2 - \frac{n(n+1)(n+2)}{3!}x^3 + \dots$$

$$\boxed{T_{r+1} = \frac{n(n-1)(n-2)\dots(n-r+1)}{r!}x^r} \sim \text{General Term!}$$

□ If n is negative integer or fraction.

$$(a+b)^n = \begin{cases} a^n(1+\frac{b}{a})^n & \text{if } |\frac{b}{a}| < 1 \\ b^n(1+\frac{a}{b})^n & \text{if } |\frac{a}{b}| < 1 \end{cases}$$

Example 1 $\Rightarrow (3+2x)^{-10}$ $T_{r+1} = (-1)^{m+r-1} {}^m C_r x^r \sim \text{General term for } (1+x)^{-m}$

$\Rightarrow (3(1+\frac{2x}{3}))^{-10}$ $T_{r+1} = {}^{m+r-1} C_r x^r \sim \text{General term for } (1-x)^{-m}$

$\Rightarrow 3^{-10} (1+\frac{2x}{3})^{-10}$

$\Rightarrow 3^{-10} \left[1 - 10(\frac{2x}{3}) + \frac{(-10)(-10+1)}{2!}(\frac{2x}{3})^2 + \dots \right]$

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10. Find range of x for which $\frac{1}{\sqrt{5+4x}}$ can be expanded in ascending Power of x

Sol: $\Rightarrow (5+4x)^{-\frac{1}{2}}$

$\Rightarrow 5^{-\frac{1}{2}} [1+4x]^{-\frac{1}{2}}$

$\Rightarrow 5^{-\frac{1}{2}} \left[1 - \frac{1}{2}(\frac{4x}{5}) + \frac{(-\frac{1}{2})(-\frac{1}{2}-1)}{2!}(\frac{4x}{5})^2 - \dots \right]$

* $(\sqrt[n]{P+Q}) = I+f$ I is integer f is fractional part $0 < f < 1$

if n is odd then which of the following is correct?

(A) I is even (C) $(I+f)f = 1$ [JEE Adv 2023]

(B) I is odd (D) $(I+f)(1-f) = 1$

$$(\sqrt[n]{P+Q}) = I+f$$

$$(\sqrt[5]{5+2}) = _ \quad 0 < f < 1$$

$$0 < f' < 1$$

$$f+f' = \text{Integer}$$

$$f+f' = 1 \quad (\because 0 < f+f' < 2) \quad 1 \text{ is odd, } f' = 1-f$$

"Note": Fractional part of $\frac{(n+1)^x}{n} = \frac{1}{n}$.

* * P.T.Q.

* * * Q. Coeff x^{203} in $(x-1)(x-2)(x-3) \dots (x-20)$ is K then Sum of digits of K is _.

• Max Possibility of x Power is $\frac{n(n+1)}{2} = \frac{20(20+1)}{2} = 210$

• Hence x^{210} is Max Power

⇒ Take x^{210} highest Power common and get the relation. As follows

$$= x^{210} \left(1 - \frac{1}{x}\right) \left(1 - \frac{2}{x^2}\right) \left(1 - \frac{3}{x^3}\right) \left(1 - \frac{4}{x^4}\right) \left(1 - \frac{5}{x^5}\right) \dots \left(1 - \frac{20}{x^{20}}\right)$$

⇒ Go for requirement as we need x^{203}

⇒ Hence we should get x^7 one side.

$$= x^{210} \left[\frac{1}{x^7} \right]$$

(-7)	$\left(\frac{1}{x^7}\right)$	$7=7$	} All Possible cases
$(-1)(-2)(-4)$	$\left(\frac{1}{x}\right)\left(\frac{1}{x^2}\right)\left(\frac{1}{x^4}\right)$	$1+2+4=7$	
$(-1)(-6)$	$\left(\frac{1}{x}\right)\left(\frac{1}{x^6}\right)$	$1+6=7$	
$(-2)(-5)$	$\left(\frac{1}{x^2}\right)\left(\frac{1}{x^5}\right)$	$2+5=7$	
$(-3)(-4)$	$\left(\frac{1}{x^3}\right)\left(\frac{1}{x^4}\right)$	$3+4=7$	

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⇒ Split in All possible cases & check coeff

⇒ Add them all